February 03, 2011

MDNRE, WRD-PSM, Grand Rapids
Attn: Ms. Chris Little
littlec@michigan.gov
Lansing, MI 48909

Project: Wastewater Surveys

Dear Ms. Chris Little,

Enclosed is a copy of the laboratory report, comprised of the following work order(s), for test samples received by TriMatrix Laboratories:

Work Order	Received	Description
1101257	01/25/2011	Cadillac WWTP, MI0020257

This report relates only to the sample(s), as received. Test results are in compliance with the requirements of the National Environmental Laboratory Accreditation Conference (NELAC). Any qualifications of results, including sample acceptance requirements, are explained in the Statement of Data Qualifications.

Estimates of analytical uncertainties for the test results contained within this report are available upon request.

If you have any questions or require further information, please do not hesitate to contact me.

Sincerely,

Lisa M. Harvey
Project Chemist

Enclosures(s)

5560 Corporate Exchange Court SE ♦ Grand Rapids, MI 49512 ♦ 616.975.4500 ♦ Fax 616.942.7463 ♦ www.trimatrixlabs.com

ANALYTICAL REPORT

Client:	**MDNRE, WRD-PSM, Grand Rapids**	Work Order:	**1101257**
Project:	Wastewater Surveys	Description:	Cadillac WWTP, MI0020257
Client Sample ID:	**CD-001A**	Sampled:	01/25/11 10:30
Lab Sample ID:	**1101257-01**	Sampled By:	C.Little
Matrix:	Waste Water	Received:	01/25/11 11:45

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

Analyte	Analytical Result		RL	MDL	Unit	Dilution Factor	Method	Date Time Analyzed	By	QC Batch
BOD, Carbonaceous (5-Day)	**1.7**	**J**	2.0	1.0	mg/L	1	SM 5210 B 20th	01/26/11 11:00	CAM	1100777

5560 Corporate Exchange Court SE ♦ Grand Rapids, MI 49512 ♦ 616.975.4500 ♦ Fax 616.942.7463 ♦ www.trimatrixlabs.com

TRIMATRIX LABORATORIES

ANALYTICAL REPORT

Client:	**MDNRE, WRD-PSM, Grand Rapids**	Work Order:	**1101257**
Project:	Wastewater Surveys	Description:	Cadillac WWTP, MI0020257
Client Sample ID:	**CD-001A-F**	Sampled:	01/25/11 07:00
Lab Sample ID:	**1101257-02**	Sampled By:	C.Little
Matrix:	Waste Water	Received:	01/25/11 11:45

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

Analyte	Analytical Result	RL	MDL	Unit	Dilution Factor	Method	Date Time Analyzed	By	QC Batch
BOD, Carbonaceous (5-Day)	**2.1**	2.0	1.0	mg/L	1	SM 5210 B 20th	01/26/11 11:00	CAM	1100777

5560 Corporate Exchange Court SE ♦ Grand Rapids, MI 49512 ♦ 616.975.4500 ♦ Fax 616.942.7463 ♦ www.trimatrixlabs.com

TRIMATRIX LABORATORIES

QUALITY CONTROL REPORT

Physical/Chemical Parameters by EPA/APHA/ASTM Methods

Analyte: BOD, Carbonaceous (5-Day)/SM 5210 B 20th

QC Batch: 1100777 (General Inorganic Prep) Analyzed: 01/26/2011 By: CAM

QC Type	Sample Conc.	Spike Qty.	Result	Unit	Spike % Rec.	Control Limits	RPD	RPD Limits	RL	MDL
Method Blank			2.0 U	mg/L					2.0	1.0
Laboratory Control Sample		198	**206**	mg/L	104	85-115		20	2.0	1.0

5560 Corporate Exchange Court SE ♦ Grand Rapids, MI 49512 ♦ 616.975.4500 ♦ Fax 616.942.7463 ♦ www.trimatrixlabs.com

STATEMENT OF DATA QUALIFICATIONS

All analyses have been validated and comply with our Quality Control Program.
No Qualifications required.

5560 Corporate Exchange Court SE ♦ Grand Rapids, MI 49512 ♦ 616.975.4500 ♦ Fax 616.942.7463 ♦ www.trimatrixlabs.com

#08091

27.5

E-1101257

LAB ORDER #

****NPDES ONLY****

MICHIGAN DEPT. OF NATURAL RESOURCES AND ENVIRONMENT
ENVIRONMENTAL LABORATORY
ANALYSIS REQUEST SHEET
TriMatrix Labs

Water

MATRIX=WASTEWATER

ACCEPT HT CODES? Y / N
If yes which parameters?

SITE CODE NUMBER	SITE NAME		
MI 0020257	Cadillac WWTP		

DIVISION	DISTRICT/OFFICE	MDNRE PROJECT MANAGER	E-MAIL ADDRESS	PHONE
WRD-PSM-Grand Rapids		Chris C. Little	littlec@michigan.gov	cell 616-550-7668

PRIMARY CONTACT PERSON	CONTRACT FIRM NAME (if applicable)	PHONE		
Chris C. Little		cell 616-550-7668	AY: 11 INDEX: 63206 PROJECT 481000	PCA: 41053 PH: 01

OVERFLOW LAB (Required)
1ST CHOICE:
2ND CHOICE:

E-MAIL ADDRESSES TO SEND ADDITIONAL REPORTS TO:
1.)
2.)

COLLECTED BY: Chris C. Little
PHONE: cell 616-550-7668

**** SAFETY INFORMATION REQUIRED ****
SEE BACK OF FORM

LAB USE ONLY		FIELD ID (Sample Identification)	SAMPLE COLLECTED DATE MM/DD/YY	SAMPLE COLLECTED TIME MILITARY	COMMENTS	
1	AB-01	CD-001A	011125	1030	No Cl2	B
2	AB-02	CD-001A-F	011125	0700	↓	B
3	AB					
4	AB					
5	AB					
6	AB					
7	AB					
8	AB					
9	AB					
10	AB					

GENERAL CHEMISTRY

BOD Carb 5 day 1 2 3 4 5 6 7 8 9 10

~~*BOD 5 day 1 2 3 4 5 6 7 8 9 10~~

Chain-of-Custody

RELEASED BY / ORGANIZATION	RECEIVED BY / ORGANIZATION	DATE	TIME
Print Name & Organization: Chris C. Little-MDNRE/RM/WRD/PSM/GR Signature: Chris C. Little	Print Name & Organization Signature: [illegible]	1/25/11	1145
Print Name & Organization Signature	Print Name & Organization Signature		
Print Name & Organization Signature	Print Name & Organization Signature		

Revised January 2010

SAMPLE RECEIVING / LOG-IN CHECKLIST

TRIMATRIX LABORATORIES

Client	MI Dept. Env.
Work Order #:	1101257
New / Add To	
Receipt Record Page/Line #	27.5
Project Chemist	JWW
Sample #s	01 - 02

Recorded by (Initials/date): WC 1/25/11

☑ Cooler ☐ Box ☐ Other______ Qty Received: 1

Thermometer Used: ☑ IR Gun (#202) ☐ Digital Thermometer (#54) ☐ Other (#______) ☐ See Additional Cooler Information Form

	Cooler 1	Cooler 2	Cooler 3	Cooler 4
Cooler #	—			
Time	1145			
Custody Seals:	☑ None ☐ Present / Intact ☐ Present / Not Intact	☐ None ☐ Present / Intact ☐ Present / Not Intact	☐ None ☐ Present / Intact ☐ Present / Not Intact	☐ None ☐ Present / Intact ☐ Present / Not Intact
Coolant Location:	Dispersed / Top / Middle / Bottom (Dispersed circled)	Dispersed / Top / Middle / Bottom	Dispersed / Top / Middle / Bottom	Dispersed / Top / Middle / Bottom
Coolant/Temperature Taken Via:	☐ Loose Ice / Avg 2-3 containers ☐ Bagged Ice / Avg 2-3 containers ☑ Blue Ice / Avg 2-3 containers ☐ None / Avg 2-3 containers	☐ Loose Ice / Avg 2-3 containers ☐ Bagged Ice / Avg 2-3 containers ☐ Blue Ice / Avg 2-3 containers ☐ None / Avg 2-3 containers	☐ Loose Ice / Avg 2-3 containers ☐ Bagged Ice / Avg 2-3 containers ☐ Blue Ice / Avg 2-3 containers ☐ None / Avg 2-3 containers	☐ Loose Ice / Avg 2-3 containers ☐ Bagged Ice / Avg 2-3 containers ☐ Blue Ice / Avg 2-3 containers ☐ None / Avg 2-3 containers
Alternate Temperature Taken Via:	☐ Temperature Blank (TB) ☐ 1 Container	☐ Temperature Blank (TB) ☐ 1 Container	☐ Temperature Blank (TB) ☐ 1 Container	☐ Temperature Blank (TB) ☐ 1 Container

Cooler 1 temperatures:

	Recorded °C	Correction Factor °C	Actual °C
Temp Blank:			
TB location: Representative / Not Representative			
1	5.7		5.7
2	5.4		5.6
3			
Average °C			5.7

Coolers 2–4: Recorded °C / Correction Factor °C / Actual °C; Temp Blank; TB location: Representative / Not Representative; 1; 2; 3; Average °C — blank.

☐ Cooler ID on COC? ☐ VOC Trip Blank received? (each cooler)

If any shaded areas checked, complete Sample Receiving Non-Conformance Form

Paperwork Received

☐ No COC Received

N/A	Yes	No	
	☑	☐	Chain of Custody record(s)?
	☑		If No, COC Initiated By ______
	☑		Rec'd for Lab Signed/Date/Time?
☑	☐		Shipping document?
☑	☐		Other ______

COC ID #s

☐ TriMatrix

☑ Other (Name or ID#)______

Check COC for Accuracy

☐ No analysis requested

Yes	No	
☑	☐	Sample ID matches COC?
☑	☐	Sample Date and Time matches COC?
☑		Container type completed on COC?
☑	☐	All container types indicated are received?

Sample Condition Summary

☐ Non-TriMatrix containers, see Notes

N/A	Yes	No	
	☐	☑	Broken containers/lids?
	☐	☑	Missing or incomplete labels?
	☐	☑	Illegible information on labels?
	☐	☑	Low volume received?
	☐	☑	Inappropriate containers received?
☑	☐	☐	VOC vials / TOX containers have headspace?
☐	☐	☑	Extra sample locations / containers not listed on COC?

Check Sample Preservation

N/A	Yes	No	
	☑	☐	Average sample temperature ≤6° C?
☐	☑		Completed Sample Preservation Verification Form?
☐	☑	☐	Samples preserved correctly?
☑	☐		If "No", added orange tag?
☑	☐		Received pre-preserved VOC soils? ☐ MeOH ☐ Na_2SO_4

Check for Short Hold-Time Prep/Analyses

☐ Bacteriological
☐ Air Bags
☐ EnCores / Methanol Pre-Preserved
☐ Formaldehyde/Aldehyde
☑ Green-tagged containers
☐ Yellow/White-tagged 1L ambers (SV Prep-Lab)

AFTER HOURS ONLY: COPIES OF COC TO LAB AREA(S)
☐ NONE RECEIVED
☑ RECEIVED, COCs TO LAB(S)

Notes

☐ Trip Blank received ☐ Trip Blank not listed on COC
☐ No COC received, Proj. Chemist reviewed (Init/Date) ______
☐ No analysis requested, Proj. Chemist completed (Init/Date) ______

Cooler Received (Date/Time)	Paperwork Delivered (Date/Time)	≤1 Hour Goal Met?
1/25/11 1145	1/25/11 1150	Yes / No (Yes circled)

TRIMATRIX LABORATORIES

SAMPLE PRESERVATION VERIFICATION FORM

Client: MI Dept Nat & Env.	Work Order #: 1101257
Receipt Log #: 27.5	Completed By (initials/date): MC 1/25/11
Project Chemist: VMH	

COC ID # 08091		Adjusted by: ____ Date: ____			DO NOT ADJUST pH FOR THESE CONTAINER TYPES			
Container Type	5	4	13		3	6	15	
Tag Color	Lt. Blue	Blue	Brown		Green	Red	Red Stripe	
Preservative	NaOH	H_2SO_4	H_2SO_4		None	HNO_3	HNO_3	
Expected pH	>12	<2	<2		6-8	<2	<2	
COC Line #1					✓			
COC Line #2					✓			
COC Line #3								
COC Line #4								
COC Line #5								
COC Line #6								
COC Line #7								
COC Line #8								
COC Line #9								
COC Line #10								

Comments

Ph Strip Lot #
☒ HC075211
☐ ____

Aqueous Samples: For each sample and container type, check the box if pH is acceptable. If pH is not acceptable for any sample container, record pH in box, and note on Sample Receiving Checklist and on Sample Receiving Non-Conformance Form. If approved by Project Chemist, add acid or base to the sample to achieve the correct pH. Add up to, but do not exceed 2x the volume initially added at container prep (see table below for initial volumes used). Add orange pH tag to sample container and record information requested. Record adjusted pH on this form. Do not adjust pH for container types 3, 6, and 15.

COC ID #		Adjusted by: ____ Date: ____			DO NOT ADJUST pH FOR THESE CONTAINER TYPES			
Container Type	5	4	13		3	6	15	
Tag Color	Lt. Blue	Blue	Brown		Green	Red	Red Stripe	
Preservative	NaOH	H_2SO_4	H_2SO_4		None	HNO_3	HNO_3	
Expected pH	>12	<2	<2		~7	<2	<2	
COC Line #1								
COC Line #2								
COC Line #3								
COC Line #4								
COC Line #5								
COC Line #6								
COC Line #7								
COC Line #8								
COC Line #9								
COC Line #10								

Comments

Container Size (mL)	Original Vol. of Preservative (mL)
Container Type 5	NaOH
500	2.5
1000	5.0
Container Type 4	H_2SO_4
125	0.5
250	1.0
500	2.0
1000	4.0
Container Type 13	H_2SO_4
500	2.5